

must have strong fundamentals in analogue, digital and mixed-signal designs. This includes a good understanding of logic design and VLSI design concepts such as RTL and gate-level verification, synthesis, place-and-route, design for test, signal integrity and power analysis. In languages, the candidates must be familiar with HDL (Verilog, VHDL, SystemVerilog), C and C++."

VLSI design is an upcoming trend in India. "We look for chip designers in two areas: IC design and system design. The front-end work on IC requires traditional HDL-based designing. We look for candidates who know HDL Verilog languages as SystemVerilog is the upcoming method. For back-end processes, candidates must be familiar with foundries, their processes and EDA tools. They require skills in verification, like use of different tools and high-level verification, behavioural and post-verification skills. In system design, they should be able to put together applications. This requires skills in board design, hardware design, exact interface protocols, and domain expertise," adds Rajesh Choudhary, head-HR, Xilinx India.

Opportunities for freshers

The chip design industry provides hardworking and competent freshers with ample scope to learn.

Texas Instruments recruits new college graduates every year. "Our campus hiring process is the foundation of our talent strategy and this has not changed over the years. We also have a strong internship programme which gives the students an opportunity to spend quality time with our talented

"We expect the candidates to have a good knowledge of the VLSI design flow. From the technical perspective, they must have strong fundamentals in analogue, digital and mixed-signal designs — Sunil Pathak, director-HR, Cadence Design Systems"



engineers in core semiconductor technology and interact with them closely," avers Bhan.

What the students learn during their internship helps them to make good decisions when they start their career in designing chips or complex systems using semiconductors.

"After the students join us, we spend several months honing their skills so that they are effective in their first job. We also associate them with experienced engineers who teach them on the job. In fact, at TI, a majority of our leaders are those whom we hired straight from engineering colleges," adds Bhan.

Internship advice

Getting hands-on experience in solving a problem bridges the gap between theory and practice. So take your internship and project work seriously.

"TI has robust internship programmes in select colleges in India. We have provided various types of tools and kits to students. Our technical talent teams also serve as visiting faculty members to universities. Additionally, we have joint development programmes with some universities where the students get to work on a cutting-edge technology research project," shares Bhan.

Cadence offers internship pro-

grammes for VLSI design students where they work alongside senior engineers on customer projects and tapeouts. In addition to technical skills, students get to learn soft skills like communications, problem-solving and multi-tasking. These internship programmes also provide an opportunity to identify potential candidates for upcoming positions. As the potential candidate is trained in the technology and has built up relationships internally and externally, the ramp to productivity is shortened.

Xilinx India, which recruits from NITs and IITs across India, has built relationships with universities like IIT in Hyderabad. It offers internships and also plans campus placements in the future.

Learning beyond the textbook

According to design firms, there exists a significant gap between the curriculum of engineering colleges and the requirements of the industry. Also, many engineering colleges are not equipped with the latest toolkits and technologies—the fact that reinforces the need for an industry-academia interface.

It is possible that you are aware of most of the technologies from a notional perspective only. But keeping in mind the industry needs, you need the actual hands-on experience. As there are a number of directions for you while choosing a career in chip designing, be clear about your goals. Dig all the possibilities to gain practical exposure and stay up-to-date on the latest technological trends to get an edge over your competitors. ●

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